

FL Simulation Results

File: test_B4

Federated Learning Configuration

Total Clients (N): 10
Malicious Clients (M): 8
Neural Network: SimpleNN
Training Rounds: 15
Poisoned Rounds (R): 15
Distribution Strategy: last
Local Epochs (E): 3

Data Poisoning Attack Parameters

Poisoning Operation: Label Flip
Attack Intensity: 10.0% (0.10)
Poisoned Data Percentage: 25.0% (0.25)

Attack Summary: Using label_flip with 10.0% (0.10) intensity on 25.0% (0.25) of data for 15 rounds with 8 malicious clients.

Simulation Results

Init Accuracy: 0.7928
Clean Accuracy: 0.8757
Clean + DP Protection Accuracy: 0.8759
Poisoned Accuracy: 0.8414
Poisoned + DP Protection Accuracy: 0.8405
Drop (Clean - Poisoned): 0.0344
Drop (Clean - Init): 0.0829
Drop (Clean DP - Init): 0.0831
Drop (Poisoned - Init): 0.0486
Drop (Poisoned DP - Init): 0.0477
DP Protection Method: foolsgold
GPU Used: GPU 0

Confusion Matrix Metrics (Weighted Avg):

Clean Precision: 0.8816
Clean Recall: 0.8757
Clean F1 Score: 0.8749

Clean DP Precision: 0.8817
Clean DP Recall: 0.8759
Clean DP F1 Score: 0.8751

Poisoned Precision: 0.8489
Poisoned Recall: 0.8414
Poisoned F1 Score: 0.8400

DP Protection Precision: 0.8479
DP Protection Recall: 0.8405
DP Protection F1 Score: 0.8389

Summary

FL Simulation Complete
Task: 11aefc89-58f5-46ed-8112-6767e35e36de
GPU: GPU 0
Init Accuracy: 0.7928
Clean Accuracy: 0.8757
Clean DP Accuracy: 0.8759
Poisoned Accuracy: 0.8414
Data Poison Protection Accuracy: 0.8405
Drop (Clean - Poisoned): 0.0344
Drop (Clean - Init): 0.0829
Drop (Clean DP - Init): 0.0831
Drop (Poisoned - Init): 0.0486
Drop (Poisoned_DP - Init): 0.0477
Data Poison Protection Method: foolsgold
--- Confusion Matrix Metrics (Weighted Avg) ---
Clean Precision: 0.8816
Clean Recall: 0.8757
Clean F1 Score: 0.8749
Clean DP Precision: 0.8817
Clean DP Recall: 0.8759
Clean DP F1 Score: 0.8751
Poisoned Precision: 0.8489
Poisoned Recall: 0.8414
Poisoned F1 Score: 0.8400
DP Protection Precision: 0.8479
DP Protection Recall: 0.8405
DP Protection F1 Score: 0.8389